MAT 125: DISCRETE MATHEMATICS II

REVISION QUESTIONS AND ANSWERS

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1. Define the following terms:

   a) Logic **[1 mark]**

   Solution:

   Logic is commonly known as the science of reasoning. ✓

   b) A contingent **[1 mark]**

2. Let p: Jitendra is a computer teacher

   q: Rahul is a student.

   Write each of the following statement in symbolic form:

   i) Jitendra is a computer teacher and Rahul is a student. **[1 mark]**

   Solution: $p \wedge q$ ✓

   ii) Jitendra is a computer teacher and Rahul is not a student. **[1 mark]**

   Solution: $p \wedge \sim q$ ✓

   iii) Jitendra is not a computer teacher but Rahul is a student. **[1 mark]**

   Solution: $\sim p \wedge q$ ✓

   iv) Neither Jitendra is a computer teacher nor Rahul is a student. **[1 mark]**

   Solution: $\sim p \wedge \sim q$ ✓

   v) Either Jitendra is a computer teacher or Rahul is a student. **[1 mark]**

   Solution: $p \vee q$ ✓

3. Write the following English sentences in symbolic form:

a) Lisa cannot play both soccer and tennis this year. **[1 mark]**

Solution: Let

s : "Lena can play soccer this year"

t : "Lena can play tennis this year"

Answer: $\sim (s \wedge t) \equiv \sim s \vee \sim t$ ✓

b) He is poor but honest. **[1 mark]**

Solution: Let

p : He is poor

q : He is honest

Then we have : $p \wedge q$ ✓

c) Neither it is hot nor cold today. **[1 mark]**

Solution: Let

p : It is hot today

q : It is cold today

So the symbolic form is $\sim p \wedge \sim q$. ✓

d) I will dance only if you sing. **[1 mark]**

Solution: We have,

The given sentence is "I will dance only if you sing."

This sentence is of the form: "p only if q".

So, the symbolic form is $p \rightarrow q$ ✓

where-

p : I will dance

q : You sing

e) Presence of cycle in a single instance RAG is a necessary and sufficient condition for deadlock. **[1 mark]**

Solution: We have-

The given sentence is- "Presence of cycle in a single instance RAG is a necessary and sufficient condition for deadlock."

This sentence is of the form- "p is necessary and sufficient for q".

So, the symbolic form is $p \leftrightarrow q$ ✓

where-

p : Presence of cycle in a single instance RAG

q : Presence of deadlock

f) Presence of cycle in a multi instance RAG is a necessary but not sufficient condition for deadlock. **[1 mark]**

Solution: We have-

The given sentence is- "Presence of cycle in a multi instance RAG is a necessary but not sufficient condition for deadlock."

This sentence is of the form- "p is necessary but not sufficient for q".

So, the symbolic form is $(q \rightarrow p) \wedge \sim (p \rightarrow q)$ ✓

where-

p : Presence of cycle in a multi instance RAG

q : Presence of deadlock

4. Negate the following:

a) $p \vee \sim q$ **[1 mark]**

Solution: $\sim (p \vee \sim q) \equiv \sim p \wedge \sim (\sim q) \equiv \sim p \wedge q$ ✓

b) $\sim p \rightarrow q$ **[1 mark]**

Solution: $p \rightarrow q \equiv \sim p \vee q$

$\sim p \rightarrow q \equiv (\sim) \sim p \vee q \equiv p \vee q$

$\sim [\sim p \rightarrow q] \equiv \sim [(\sim) \sim p \vee q] \equiv \sim [p \vee q] \equiv \sim p \wedge \sim q$ ✓

c) $p \leftrightarrow q$ **[1 mark]**

Solution: $p \leftrightarrow q \equiv p \oplus q$ ✓

d) $p \rightarrow (q \wedge r)$ **[1 mark]**

Solution: $\sim [\, p \rightarrow (q \wedge r)\,] \equiv \sim\ [\, \sim p\ \vee\ (q\ \wedge\ r)\,] \equiv p \wedge\ \sim (q\ \wedge\ r)$ ✓

5. In the conditional statement $p \rightarrow q$, what term is given to the variables p and q respectively? **[2 marks]**

Solution:

- p is called the premise, hypothesis, or the antecedent. ✓
- q is called the conclusion or consequent. ✓

6. Write the statement:"To be a citizen of this country, it is sufficient that you were born in Kenya" in the form "if p, then q " in English. **[2 marks]**

Solution: p is sufficient for q means "if p, then q". ✓

Hence the statement is:

"If you are a citizen of this country, then you were born in Kenya" ✓

7. Given the implication: If this book is interesting, then I am staying at home, write its converse, inverse and Contrapositive. **[3 marks]**

Solution:

Converse: If I am staying at home, then this book is interesting. ✓

Inverse: If this book is not interesting, then I am not staying at home. ✓

Contrapositive: If I am not staying at home, then this book is not interesting. ✓

8. Formalize in symbols the statement "For every positive integer x, there exists a positive integer y such that y is greater than x." **[2 marks]**

Solution:

$\forall x \in \mathbb{Z}^+ \quad \exists y \in \mathbb{Z}^+\ (y > x)$. ✓✓

9. Translate the following formalized statement into English:

$\exists y(\text{Parent}(y, \text{John}) \wedge \neg\text{Happy}(y))$." **[2 marks]**

Solution:

There exists a person y who is a parent of John and is not happy. ✓✓

10. Simplify the Boolean expression $F(X,Y,Z) = X'Y + YZ' + YZ + X'YZ'$. **[5marks]**

Solution:

$$
\begin{aligned}
F(X,Y,Z) &= X'Y + YZ' + YZ + X'YZ' \quad \text{(Given expression)} \\
&= X'Y + Y(Z' + Z) + X'YZ' \quad \text{(Distributive property)} \checkmark \\
&= X'Y + Y + X'YZ' \quad \text{(Complement property: } Z' + Z = 1) \checkmark \\
&= Y + X'Y + XY'Z' \quad \text{(Commutative property)} \\
&= Y + X'Y(1 + Z') \quad \text{(Distributive property)} \checkmark \\
&= Y + X'Y \quad \text{(Complement property: } 1 + Z' = 1) \checkmark \\
&= Y(1 + X') \quad \text{(Distributive property)} \\
&= Y \quad \text{(Complement property: } 1 + X' = 1) \checkmark
\end{aligned}
$$

11. Without constructing the truth table show that $p \to (q \to p) \equiv t$, where t denotes a tautology. **[3 marks]**

Solution:

$$
\begin{aligned}
p \to (q \to p) &\equiv \sim p \ \vee \ (\sim q \ \vee \ p) \\
&\equiv \sim p \ \vee \ p \ \sim q \\
&\equiv t \vee \ \sim q \\
&\equiv t
\end{aligned}
$$

12. Using a truth table , prove the following relation: $(P \wedge Q) \vee (P \wedge \sim Q) \equiv P$ **[4 marks]**

Solution:

Constructing the truth table:

P	Q	$\sim Q$	$(P \wedge Q)$	$(P \wedge \sim Q)$	$(P \wedge Q) \vee (P \wedge \sim Q)$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	F	F	F
F	F	T	F	F	F
			✓	✓	✓

Now, let's compare the truth values of $(P \wedge Q) \vee (P \wedge \sim Q)$ and P for each row. As we can see, the two expressions are equal for all possible combinations of truth values for P and Q. ✓

Therefore, we have proved that $(P \wedge Q) \vee (P \wedge \sim Q) \equiv P$ using the truth table.

13. Define the following terms: a tautology a contradiction and a contingency. **[3 marks]**

Solution:

A tautology is a logical statement that is always true, regardless of the truth values of its individual components. ✓

A contradiction is a logical statement that is always false, regardless of the truth values of its individual components. ✓

A contingent statement is a logical statement that is neither always true (tautology) nor always false (contradiction). It is true for some combinations of truth values of its variables and false for others. ✓

14. Let $P(x)$ be the statement "the word x contains the letter a." What are the truth values of?

i) P(true). **[1 mark]**

Solution:

ii) P(false). **[1 mark]**

Solution:

15. Let t denote a tautology, F the truth value FALSE , and p a proposition. Evaluate:

    i) $p \vee F$ **[1 mark]**

    Solution:

    ii) $\sim t$ **[1 mark]**

    Solution:

16. Find the bitwise OR, bitwise AND, and bitwise XOR of the following pairs of bit strings. $11110000, 10101010$ **[3 marks]**

    Solution:

17. Let p and q be the following propositions p: it is below freezing q: it is snowing Write the following propositions using p and q and logical connectives.

    i) It is below freezing but not snowing. **[1 mark]**

    Solution:

    ii) It is neither below freezing nor snowing. **[1 mark]**

    Solution:

18. Construct a truth table for the compound proposition

    $\neg(p \wedge q) \vee (\neg r)$. **[3 marks]**

    Solution:

19. Find the bounded and free variables in the following statement: **[2 marks]**

    $\exists y(x + y = 4)$

    Solution:

20. What is the scope of the existential quantifier in (b) above. **[2 marks]**

    Solution:

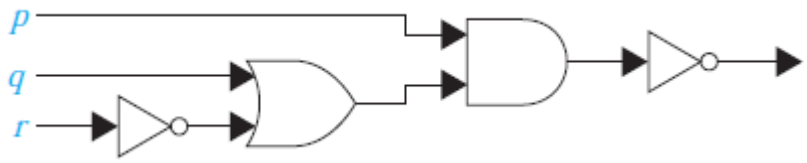
21. The function $F(x, y, z) = xy + \overline{z}$ from 3−tuples of Boolean variables to the set $\{0, 1\}$ is a Boolean function of deg 3.

    Display this function in an input output table. **[3 marks]**

Solution:

22. Find the output of the following combinatorial circuit: **[2 marks]**

Solution:



Solution:

23. Use Mathematical induction to show that,

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1,$$

for all non-negative integers n **[4 marks]**

Solution:

24. Construct a truth table for the exclusive or of p and q denoted by $p \oplus q$. **[2 marks]**

Solution:

25. Simplify the Boolean expression: $abc' + ac'$. **[2 marks]**

Solution:

$abc' + ac' = ac(b' + 1) = ac$ ✓✓

26. Find the complement of the Boolean expression $A'B + CD'$, **[2 marks]**

Solution: Apply De Morgan's laws:

$(A'B + CD')' = (A'B)' \cdot (CD')' = (A + B') \cdot (C' + D)$ ✓✓

27. Show that if an integer n is not divisible by 3, then $n^2 - 1$ must be a multiple of 3. **[5 marks]**

Solution:

Assume that n is not divisible by 3 . This means that n leaves a remainder of 1 or 2 when divided by 3. ✓

Case 1: n leaves a remainder of 1 when divided by 3.

In this case, $n = 3k + 1$ for some integer k. ✓

Then $n^2 - 1 = (3k + 1)^2 - 1 = 9k^2 + 6k + 1 - 1 = 3(3k^2 + 2k)$ ✓

So, $n^2 - 1$ is clearly divisible by 3.

Case 2: n leaves a remainder of 2 when divided by 3.

In this case, $n = 3k + 2$ for some integer k. ✓

Then $n^2 - 1 = (3k + 2)^2 - 1 = 9k^2 + 12k + 4 - 1 = 3(3k^2 + 4k + 1)$ ✓

So, $n^2 - 1$ is again divisible by 3.

In both cases, we have shown that if n is not divisible by 3, then $n^2 - 1$ is a multiple of 3. Therefore, the original statement holds true.

28. Prove the statement: For all integers m and n, if m and n are odd integers, then $m + n$ is an even integer. **[3 marks]**

Solution:

29. Consider the statements:

p: You work hard.

q: You will succeed in your life.

Translate each of the following symbolic expressions into meaningful sentences:

(i) $p \Rightarrow q$ **[1 mark]**

Solution:

If you work hard then you will succeed in your life.

(ii) $q \Rightarrow p$ **[1 mark]**

Solution:

If you will succeed in your life it means that you work hard.

(iii)) $(\sim p) \Rightarrow (\sim q)$ **[1 mark]**

Solution:

If you don't work hard then you will not succeed in your life.

(d) $(\sim q) \Rightarrow (\sim p)$ **[1 mark]**

Solution:

If you don't succeed in your life it means that you didn't work hard.

30. How many rows appear in the truth table of this compound propositions?

$(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \sim t)$ **[2 marks]**

Solution:

To determine the number of rows in the truth table for the compound proposition $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \sim t)$, we need to consider all possible combinations of truth values for the variables p, q, r, s, and t.

If there are n variables, each with two possible truth values (True or False), then the number of rows in the truth table is 2^n. ✓

In this case, we have five variables (p, q, r, s, t), so the number of rows is $2^5 = 32$. ✓

Therefore, the truth table for the given compound proposition will have 32 rows.

31. Determine the number of rows in the truth table for the compound proposition $(a \wedge b \wedge c) \vee (\sim d \wedge e) \vee (f \wedge \sim g)$. **[2 marks]**

Solution:

To find the number of rows in the truth table for the compound proposition $(a \wedge b \wedge c) \vee (\sim d \wedge e) \vee (f \wedge \sim g)$, we need to consider all possible combinations of truth values for the variables a, b, c, d, e, f, and g.

If there are n variables, each with two possible truth values (True or False), then the number of rows in the truth table is 2^n. ✓

In this case, we have seven variables (a, b, c, d, e, f, g), so the number of rows is $2^7 = 128$. ✓

Therefore, the truth table for the given compound proposition will have 128 rows.

32. Write the converse, inverse and contra-positive of the following statement:

" If you work hard, then you will pass" **[3 marks]**

Solution:

Let the given statement "If you work hard, then you will pass" be represented by the conditional statement $P \rightarrow Q$, where:

- P is the statement "You work hard."

- Q is the statement "You will pass."

- Conditional: "If you work hard, then you will pass." ($P \rightarrow Q$)

- Inverse: "If you do not work hard, then you will not pass." ($\sim P \rightarrow \sim Q$) ✓

- Converse: "If you will pass, then you have worked hard." ($Q \rightarrow P$) ✓

- Contrapositive: "If you will not pass, then you have not worked hard." ($\sim Q \rightarrow \sim P$) ✓

33. Write the converse, inverse and contra-positive of the following statement:

" If I run fast, then I will win the race" **[3 marks]**

Solution:

Let the given statement "If I run fast, then I will win the race" be represented by the conditional statement $P \rightarrow Q$, where:

- P is the statement "I run fast."

- Q is the statement "I will win the race."

- Conditional: "If I run fast, then I will win the race." ($P \rightarrow Q$)

- Inverse: "If I do not run fast, then I will not win the race." ($\sim P \rightarrow \sim Q$)

- Converse: "If I will win the race, then I ran fast." ($Q \rightarrow P$)

- Contrapositive: "If I will not win the race, then I did not run fast." ($\sim Q \rightarrow \sim P$)

34. Write the converse, inverse and contra-positive of the following statement:

" If it rains, then you will not play." **[3 marks]**

Solution:

Conditional: "If it rains, then you will not play." ($P \rightarrow Q$)

- Converse: "If you will not play, then it rains." ($Q \rightarrow P$) ✓

- Inverse: "If it does not rain, then you will play." ($\sim P \rightarrow \sim Q$) ✓

- Contrapositive: "If you play, then it does not rain." ($\sim Q \rightarrow \sim P$) ✓

35. Write the inverse, converse and contra-positive of the following statement " if $6+1 \geq 8$, then goats can dance" **[3 marks]**

Solution:

Let the given statement "if $6+1 \geq 8$, then goats can dance" be represented by the conditional statement $P \rightarrow Q$, where:

- P is the statement "$6+1 \geq 8$."

- Q is the statement "Goats can dance."

- Conditional: "If $6+1 \geq 8$, then goats can dance." ($P \rightarrow Q$)

- Inverse: "If $6+1 < 8$, then goats cannot dance." ($\sim P \rightarrow \sim Q$) ✓

- Converse: "If goats can dance, then $6+1 \geq 8$." ($Q \rightarrow P$) ✓

- Contrapositive: "If goats cannot dance, then $6+1 < 8$." ($\sim Q \rightarrow \sim P$) ✓

36. Using rules of inference, show that the premises $\sim p \wedge q$, $r \rightarrow p$, $\sim r \rightarrow s$ and $s \rightarrow t$ infer the conclusion t. **[4 marks]**

Solution:

To show that the conclusion t can be inferred from the given premises using rules of inference, we can use a sequence of logical steps. Here's a proof:

1. **Premise 1: $\sim p \wedge q$** (Given)

2. **Premise 2: $r \rightarrow p$** (Given)

3. **Premise 3: $\sim r \rightarrow s$** (Given)

4. **Premise 4: $s \rightarrow t$** (Given)

Now, let's use rules of inference to derive the conclusion t:

5. **Assumption: $\sim p$** (Specialization rule on premises 1) ✓

6. **Conjunction Elimination: $\sim p$** (From 1, using conjunction elimination)

7. **Modus Tollens: $\sim r$** (From 2 and 6) ✓

8. **Modus Ponens: s** (From 3 and 7) ✓

9. **Modus Ponens: t** (From 4 and 8) ✓

37. Rewrite the following quantified conditional statement informally i.e without using quantifiers or variables.

"Squareness is a sufficient condition for rectangularity." **[1 mark]**

Solution:

"If something is square, then it is also rectangular." ✓

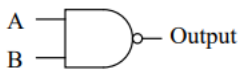
38. Rewrite the following without quantifiers or variables. **[1 mark]**

$\forall x, if x > 2$, then $x^2 > 4$

Solution:

"For any number greater than 2, its square is greater than 4." ✓

39. Identify this logic gate by name and complete the table. **[2 mark]**.



A	B	Output
0	0	
0	1	
1	0	
1	1	

Solution:

40. Write formal negations for the following statements:

i) $\forall$ primes p, p is odd. **[1 mark]**

Solution:

i) $\exists$ a triangle T such that the sum of the angles of T equals $200°$. **[1 mark]**

Solution:

41. If Socrates is human, then Socrates is mortal.

Socrates is human.

$\therefore$ Socrates is mortal.

i) Write the argument form of the above argument and determine whether it is valid. **[2 marks]**

Solution:

ii) Determine whether we can conclude that the conclusion is true if the premises are true? **[1 mark]**

Solution:

42. Show that the negation of a conditional statement p implies q is logically equivalent to p and not q. **[2 marks]**

Solution:

43. By developing a series of logical equivalences, show that

$\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p$. **[4 marks]**

Solution:

Simplify the left-hand side step by step using logical equivalences:

Apply De Morgan's Law:

$\neg(\neg p \wedge q) \wedge (p \vee q) \equiv (\neg(\neg p) \vee \neg q) \wedge (p \vee q)$ ✓

Simplify: $(p \vee \neg q) \wedge (p \vee q)$ ✓

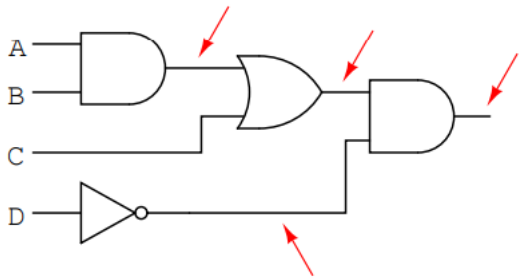
Apply Distribution:

$(p \vee \neg q) \wedge (p \vee q) \equiv p \vee (\neg q \wedge q)$ ✓

Since $\neg q \wedge q$ is always false, this simplifies to: $p \vee \text{False} \equiv p$ ✓

Therefore, by a series of logical equivalences, we have shown that $\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p$.

44. Convert the following logic gate circuit into a Boolean expression, writing Boolean sub-expressions next to each gate output in the diagram. **[3 marks]**



Solution:

45. State the rule of inference used in each of these arguments:

    i) If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed. **[1 mark]**

    Solution:

    ii) Stacy is a mathematics major and computer science major. Therefore, Stacy is a Mathematics major. **[1 mark]**

    Solution:

    iii) If I work all night on then homework, then I can answer all the exercise. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material. **[1 mark]**

    Solution:

46. Using a truth table, show that $\neg p \rightarrow (q \rightarrow r)$ is logically equivalent to $p \rightarrow (q \vee r)$. **[3 marks]**

Solution:

47. Use boolean identities to show that $x + 1 = 1$, where x is a boolean variable. **[3 marks]**

Solution:

48. Are these system specifications consistent? "The file system is not locked only if new messages will be queued ($\sim p \rightarrow q$). If the file system is not locked, then the system is functioning normally, and conversely ($\sim p \leftrightarrow r$). For new messages are not queued, it is necessary that they will be sent to the message buffer ($\sim q \rightarrow s$). Whenever the file system is not locked, new messages will be sent to the message buffer ($\sim p \rightarrow s$). New messages will not be sent to the message buffer ($\sim s$). **[4 marks]**

Solution:

49. Is $x + 4 = 2$ a proposition? Explain your answer. **[2 marks]**

Solution:

50. Prove by contraposition that if $5n + 2$ is odd then n is odd. **[3 marks]**

Solution:

51. Express the negation of the statement $\forall x \exists y(xy = 1)$ so that no negation precedes a quantifier. **[2 marks]**

Solution:

$\sim [\forall x \exists y(xy = 1)] \equiv \exists x \forall y \sim (xy = 1) \equiv \exists x \forall y(xy \neq 1)$. ✓✓

52. Construct a truth table for: $p \oplus \sim q$. **[2 marks]**

Solution:

p	**q**	$\sim$ **q**	**p** $\oplus \sim$ **q**
T	T	F	T
T	F	T	F
F	T	F	F
F	F	T	T

✓✓

53. Evaluate the expression: $(11011 \vee 01010) \wedge (10001 \vee 11011)$. **[2 marks]**

Solution:

$(11011 \vee 01010) \wedge (10001 \vee 11011) = (11011) \wedge (11011) = (11011)$ ✓✓

54. Use rules of inference to show that if the premises

$\forall x(P(x) \to Q(x)), \forall x(Q(x) \to R(x))$, and $\sim R(a)$, where a is in the domain, are true, then the conclusion $\sim P(a)$ is true. **[3 marks]**

Solution:

$\forall x(P(x) \to Q(x))$

$\forall x(Q(x) \to R(x))$

$\therefore \forall x(P(x) \to R(x))$ by universal law of syllogism ✓✓

$\forall x(P(x) \to R(x))$

$\sim R(a)$

$\sim P(a)$ by universal modus tollens ✓

55. Find free variable(s) in the following formulas:

i) $\forall x.\,(p(x) \to \exists y.\sim q\,(f(x), y, f(y)))$ **[1 mark]**

Solution: No free variable ✓

ii) $\forall x\,(\exists y.r\,(x, f(y)) \to r(x, y))$ **[1 mark]**

Solution: y free ✓

iii) $\forall z.\,(p(z) \to \exists y.\,(\exists x.q(x, y, z) \vee q(z, y, x)))$ **[1 mark]**

Solution: x free ✓

56. Consider the following logical expression:

$$\exists x(P(x) \wedge Q(y, x)) \vee \forall y\, R(z, y)$$

Determine the free variable(s) and bound variable(s) in the given expression.

Solution:

Free Variables: z (These variables are not quantified and can take any value.) ✓

Bound Variables: x, y (This variable is quantified either existentially or universally.)✓

57. Evaluate the free and bound variables in the expression:

$$\forall x \, \exists y \, (P(x, y) \wedge \neg Q(z, y))$$

Solution:

Free Variables: z (This variable is not quantified and can take any value.) ✓

Bound Variables: x, y (These variables are quantified universally or existentially.) ✓

58. Given the logical expression:

$$\exists x \, \forall y \, (P(x, y) \vee \neg Q(z)) \wedge R(w, x)$$

Identify the free and bound variables. **[2 marks]**

Solution:

Free Variables: z, w (These variables are not quantified and can take any value.) ✓

Bound Variables: x, y (These variables are quantified either existentially or universally.)✓

59. Examine the expression:

$$\forall x \, (P(x) \wedge \exists y \, (Q(y) \vee R(z, x)))$$

Find the free and bound variables.

Solution:

Free Variables: z (This variable is not quantified and can take any value.) ✓

Bound Variables: x, y (These variables are quantified universally or existentially.) ✓

60. Simplify the Boolean expression $Y = AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + B\bar{C}D$ and draw the logic circuit fort the simplified expression. **[4 marks]**

Solution:

$$\begin{aligned} Y =& AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + B\bar{C}D \\ &= B\bar{C}\bar{D}(A + \bar{A}) + B\bar{C}\bar{D} \qquad \checkmark \\ &= B\bar{C}\bar{D}(1) + B\bar{C}\bar{D} \ldots\ldots\ldots\ldots\ldots A + \bar{A} = 1 \qquad \checkmark \\ &= B\bar{C}\bar{D} + B\bar{C}\bar{D} \\ &= B\bar{C}(\bar{D} + D) \\ &= B\bar{C}(1) \\ &= B\bar{C} \qquad \checkmark \end{aligned}$$

61. Write the conclusion that can be inferred using universal modus tollens given:

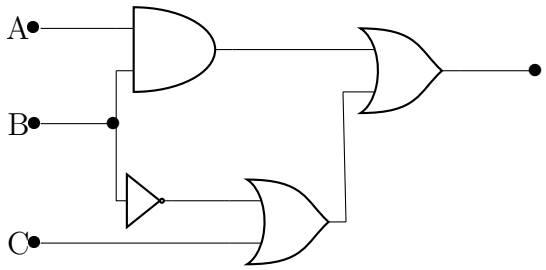
    All professors are absent-minded.

    Tom Hutchins is not absent-minded.

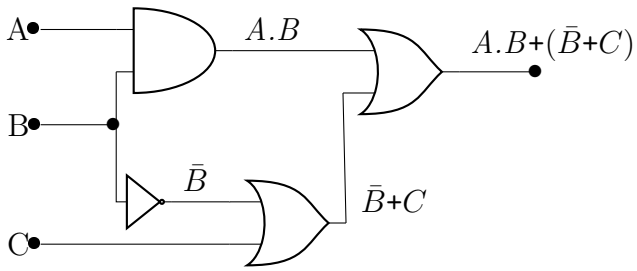
    $\therefore$ ..... **[1 mark]**

    Solution: Tom Hutchins is not a professor. ✓

62. Find the Boolean expressions that correspond to the circuit shown below. **[2 marks]**



    Solution:



63. Disprove the converse of the statement: For all integers a and b , if $a + b$ is odd, then

a is odd or b is odd. **[3 marks]**

Solution: Converse: For all integers a and b ,if a is odd or b is odd then $a + b$ is odd. ✓

Proof by counter example: ✓

Statement not true since $1 + 3 = 4$ where 1 is odd, 3 is odd but the sum is even. ✓

64. Determine whether $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$ is a tautology. **[3 marks]**

Solution:

p	**q**	$\sim$ q	p $\rightarrow$ q	$\sim$ q $\wedge$ p $\rightarrow$ q	$\sim$ p	[$\sim$q $\wedge$ (p $\rightarrow$ q)]$\rightarrow \sim$ p
T	T	F	T	F	F	T
T	F	T	F	F	F	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T

✓✓

The last column has all $T's$, hence a tautology. ✓

65. Prove the statement: For all integers m and n, if m and n are odd integers, then $m + n$ is an even integer. **[3 marks]**

Solution:

Assume m and n are odd integers .

This implies that $m = 2a + 1, n = 2b + 1, a, b \in \mathbb{Z}$ ✓

$\Rightarrow m + n = 2a + 1 + 2b + 1 = 2a + 2b + 2 = 2(a + b + 1)$ ✓

$\therefore m + n$ is even by definition of an even number ✓

66. Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics.

You learned discrete mathematics.

Therefore, you did every problem in this book. **[3 marks]**

Solution:

Let p be the proposition "You did every problem in this book". Let q be the proposition "You learned discrete mathematics". Then this argument is of the form: if

$p \to q$ and q, then p. ✓

This is an example of an incorrect argument using the fallacy of affirming the conclusion(converse error). ✓

Hence invalid. ✓

67. Show that $\sim (p \vee (\sim p \wedge q))$ and $\sim p \wedge \sim q$ are logically equivalent by developing a series of logical equivalences. **[3 marks]**

Solution:

$$\begin{aligned}
\sim (p \vee (\sim p \wedge q)) &\equiv \sim p \wedge \sim (\sim p \wedge q) \ldots\ldots\ldots\ldots\ldots\ldots \textit{by De Morgan law} \\
&\equiv \sim p \wedge [\sim (\sim p) \vee \sim q] \ldots\ldots\ldots\ldots\ldots \textit{by De Morgan law} \quad \checkmark \\
&\equiv \sim p \wedge (p \vee \sim q) \ldots\ldots\ldots\ldots\ldots\ldots\ldots \textit{by double negation law} \\
&\equiv (\sim p \wedge p) \vee (\sim p \wedge \sim q) \ldots\ldots\ldots \textit{by the distributive law} \quad \checkmark \\
&\equiv \mathbf{c} \vee (\sim p \wedge \sim q) \ldots\ldots\ldots\ldots\ldots\ldots (\sim p \wedge p) = \mathbf{c} \\
&\equiv (\sim p \wedge \sim q) \ldots\ldots\ldots\ldots\ldots\ldots\ldots \textit{by the identity law} \quad \checkmark
\end{aligned}$$

68. Determine whether the following compound statement is a tautology or contradiction

$(p \to (q \to r)) \to ((p \wedge q) \to r)$ **[5 marks]**

Solution:

p	q	r	$q \to r$	$p \to (q \to r)$	$p \wedge q$	$(p \wedge q) \to r$	$(p \to (q \to r)) \to ((p \wedge q) \to r)$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	T
T	F	T	T	T	F	T	T
T	F	F	T	T	F	T	T
F	T	T	T	T	F	T	T
F	T	F	F	T	F	T	T
F	F	T	T	T	F	T	T
F	F	F	T	T	F	T	T

A tautology

69. Determine whether $\psi = (\sim p \vee q) \wedge (q \rightarrow \sim r \wedge \sim p) \wedge (p \vee r)$ is satisfiable. **[6 marks]**

Solution:

p	q	r	$\sim p$	$\sim r$	$\sim p \vee q$	$\sim r \wedge \sim p$	$q \rightarrow \sim r \wedge \sim p$	$p \vee r$	ψ
T	T	T	F	F	T	F	F	T	F
T	T	F	F	T	T	F	F	T	F
T	F	T	F	F	F	F	T	T	F
T	F	F	F	T	F	F	T	T	F
F	T	T	T	F	T	F	F	T	F
F	T	F	T	T	T	T	T	F	F
F	F	T	T	F	T	F	T	T	T
F	F	F	T	T	T	T	T	F	F
					✓	✓	✓	✓	✓

There exists an interpretation, row 7, satisfying ψ thus ψ is satisfiable. ✓

70. Let $A = \{2, 3, 4, 5.6\}$. Determine the truth value of each of the following statements:

i) $\exists\, x \in A : x + 2 < 7$ **[1 mark]**

Solution: T since it is true for $x = 2, 3$ or 4 ✓

ii) $\forall\, x, y \in A : x + y \leq 10$ **[1 mark]**

Solution: F . $5 + 6$ is not ≤ 10. ✓

iii) $\forall\, x \in A, \exists\, y \in A : x + y \leq 10$ **[1 mark]**

Solution: T since for any number in the set, there is a number you can add and the sum will be less than 10 ✓

iv) $\exists\, x \in A, \forall\, y \in A : x + y \leq 10$ **[1 mark]**

Solution: T since 2 added to any of the element in the set has sum ≤ 10 ✓

71. Let $B = \{-2, 0, 3, 5, 7\}$. Determine the truth set for the statement

$\exists y \in B : 2y - 1 > 3$. **[2 marks]**

Solution:

The given statement can be read as "There exists an element y in the set B such that $2y - 1 > 3$." Simplifying the inequality:

$2y - 1 > 3$: Adding 1 to both sides: $2y > 4$ Dividing both sides by 2: $y > 2$. ✓

So, any element y in the set B that is greater than 2 satisfies the given inequality.

Therefore, the truth set for the statement $\exists y \in B : 2y - 1 > 3$ is $\{3, 5, 7\}$. ✓

72. Socrates says:

If I am guilty, I must be punished;

I am guilty.

Thus I must be punished.

Is the argument logically correct? **[3 marks]**

Solution: The argument is logically correct: if p means I am guilty and q means I must be punished, then: ✓

$p \to q$

p

$\therefore q$ ✓ is valid by the modus ponens inference rule ✓

73. Prove that: There are no integers x and y such that $x^2 = 4y + 2$. **[3 marks]**

Solution:

Proof by contradiction

Suppose there are integers x and y such that $x^2 = 4y + 2 = 2(2y + 1)$,so x^2 is even which in turn implies x is even. ✓

So $x = 2k,\ k \in \mathbb{Z}$. Then $x^2 = 4k^2$.

$\Rightarrow 2k^2 = 2y + 1$. This equality is not possible since the LHS is even and RHS odd.

A contradiction. ✓

So there are no integers x and $y$such that $x^2 = 4y + 2$. ✓

74. Write the negation, converse , contrapositive and the inverse of the statement:

$\forall x\ (x < 1 \Rightarrow x^2 < 1)$ **[4 marks]**

Solution:

Negation: $\exists x(x < 1 \wedge x^2 \geq 1)$. ✓

Converse: $\forall x\ (x^2 < 1 \Rightarrow x < 1)$. ✓

Contrapositive : $\forall x\ (x^2 \geq 1 \Rightarrow x \geq 1)$. ✓

Inverse : $\forall x(x \geq 1\ \Rightarrow\ x^2 \geq 1)$. ✓

75. Let t denote a tautology, c a contradiction , and p a proposition.Evaluate:

i) $p \wedge t$ **[1 mark]**

Solution: p ✓

ii) $p \wedge c$ **[1 mark]**

Solution: c ✓

76. Let p mean "Paola is happy", q mean "Paola paints a picture", and r mean "Renzo is happy'.Formalize the following sentences:

i) If Paola is happy and paints a picture then Renzo isn't happy **[1 mark]**

Solution: $p \wedge q \rightarrow \sim r$ ✓

ii) Paola is happy only if she paints a picture. **[1 mark]**

Solution: $p \rightarrow q$ ✓

77. Determine whether $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent using a truth table **[4 marks]**

Solution:

p	q	r	$p \to r$	$q \to r$	$(p \to r) \wedge (q \to r)$	$p \wedge q$	$(p \wedge q) \to r$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	F	T
F	T	F	T	F	F	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

✓ ✓ ✓

Not logically equivalent ✓

78. Proof that for every $n \in \mathbb{Z}^+$, $1^2 + 2^2 + ... + n^2 = \dfrac{n(n+1)(2n+1)}{6}$. **[5 marks]**

Solution: Proof by mathematical induction:

For every integer $n \geq 1$, let $P(n)$ be the statement

$P(n) : 1^2 + 2^2 + ... + n^2 = \dfrac{n(n+1)(2n+1)}{6}$

When $n = 1$, the statement $P(1) : 1 = \dfrac{1(1+1)(2.1+1)}{6}$ is true. ✓

Let k be an arbitrary but fixed integer such that $k \geq 1$ and assume $P(k)$ is true: that is $P(k) : 1^2 + 2^2 + ... + n^2 = \dfrac{k(k+1)(2k+1)}{6}$. ✓

We show that $P(k+1)$ is true.i.e. $1^2 + 2^2 + ... + (k+1)^2 = \dfrac{(k+1)(k+2)(2k+3)}{6}$.

Evaluating the LHS of this equation , we have,

$$\begin{aligned}
(1^2 + 2^2 + ... + (k+1)^2 &= 1^2 + 2^2 + ... + k^2) + (k+1)^2 \quad \checkmark \\
&= \frac{k(k+1) + (2k+1)}{6} + (k+1)^2 \quad \checkmark \\
&= \frac{k(k+1) + (2k+1)}{6} + \frac{6(k+1)^2}{6} \\
&= \frac{(k+1)(2k+1) + 6(k+1)}{6} \\
&= \frac{(k+1)(2k^2 + 7k + 6)}{6} \quad \checkmark \\
&= \frac{(k+1)(k+2)(2k+3)}{6}
\end{aligned}$$

79. Consider the following truth table.

			Output
A	B	C	Y
1	1	1	0
1	1	0	0
1	0	1	1
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	0

i) Determine and simplify the logic expression for the output Y above. **[3 marks]**

Solution:

There are two rows with output 1's and their corresponding binary values are 001 and 101 which correspond to product terms $\bar{A}\bar{B}C$ and $A\bar{B}C$ respectively. ✓

The resulting expression for the output is

$Y = \bar{A}\bar{B}C + A\bar{B}C = (\bar{A} + A)\bar{B}C = 1.\bar{B}C = \bar{B}C.$ ✓✓

ii) Sketch the logic circuit for the simplified expression. **[2 marks]**

Solution:

80. Find the truth set for each propositional function $p(x)$ defined on the set $\mathbb{N}$ of positive integers.

i) $p(x) : x + 2 > 7$ **[1 mark]**

Solution:

Truth set is:$\{6, 7, 8, ...\}$ ✓

ii) $p(x) : x + 5 < 3$ **[1 mark]**

Solution:

Truth set is:the empty set ✓

iii) $p(x) : x + 5 > 1$ **[1 mark]**

Solution:

Truth set is: $\mathbb{N}$ ✓

81. You are about to leave for campus in the morning and discover that you don't have your glasses. You know the following statements are true:

a. If I was reading the newspaper in the kitchen, then my glasses are on the kitchen table.

b. If my glasses are on the kitchen table, then I saw them at breakfast.

c. I did not see my glasses at breakfast.

d. I was reading the newspaper in the living room or I was reading the newspaper in the kitchen.

e. If I was reading the newspaper in the living room then my glasses are on the coffee table.

Where are the glasses? **[5 marks]**

Solution: Let

RK = I was reading the newspaper in the kitchen.

GK = My glasses are on the kitchen table.

SB = I saw my glasses at breakfast.

RL = I was reading the newspaper in the living room.

GC = My glasses are on the coffee table.

$RK \rightarrow GK$..... by (a)

$GK \rightarrow SB$..... by (d)

$\therefore RK \rightarrow SB$..... by transitivity .....(1) ✓

$RK \rightarrow SB$..... by the conclusion of (1)

$\sim SB$..... by (c)

$\therefore \sim RK$..... by modus tollens .....(2) ✓

$RL \vee RK$..... by (d)

$\sim RK$..... by the conclusion of (2)

$\therefore RL$..... by elimination .....(3) ✓

$RL \rightarrow GC$..... by (e)

RL..... by the conclusion of (3)

$\therefore GC$..... by modus ponens ✓

Thus the glasses are on the coffee table. ✓

82. Find the duals of these Boolean expressions:

i) $xyz + \bar{x}\bar{y}\bar{z}$. **[1 mark]**

Solution:

$(x + y + z)(\bar{x} + \bar{y} + \bar{z})$ ✓

ii) $x\bar{z} + x.0 + \bar{x}.1$ **[1 mark]**

Solution:

$(x + \bar{z})(x + 1)(\bar{x} + 0)$ ✓

83. De-Morganize the Boolean expression: $\overline{A(\overline{B + \overline{C}})D}$ **[2 marks]**

Solution:

$$\overline{A(\overline{B + \overline{C}})D} = \overline{A} + \overline{\overline{B + \overline{C}}} + \overline{D} \quad \checkmark$$
$$= \overline{A} + B + \overline{C} + \overline{D} \quad \checkmark$$

84. Prove that if n^2 is odd, then n is odd, where n is an integer . **[3 marks]**

Solution:

Contrapositive proof: We show that if n is even, then n^2 is even. ✓

Assume n is even, this implies $n = 2k, k \in \mathbb{Z}$ ✓

This implies $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ an even number. ✓

85. State and prove that the law of syllogism is a valid argument. **[6 marks]**

Solution:

Law: $p \to q, q \to r, \therefore\ p \to r$. ✓

p	q	r	$p \to q$	$q \to r$	$p \to r$	Critical rows
T	T	T	T	T	T	$*$
T	T	F	T	F	F	
T	F	T	F	T	T	
T	F	F	F	T	F	
F	T	T	T	T	T	$*$
F	T	F	T	F	T	
F	F	T	T	T	T	$*$
F	F	F	T	T	T	$*$

✓ ✓ ✓ ✓

All the critical rows are true hence the argument is valid. ✓

86. Use De Morgan's laws to write the negation of $4 \geq x > -1$. **[3 marks]**

Solution:

87. Determine whether the following compound proposition is a tautology and whether is satisfi

able or unsatisfiable $(p \vee q) \wedge \neg p \wedge \neg q$. **[4 marks]**

Solution:

88. Construct an identity from the absorption law $x(x + y) = x$ by taking its dual **[1 mark]**

Solution:

89. Prove that the negation of the biconditional "p if and only if q" is logically equivalent to the exclusive or disjunctive form of p and q. **[2 marks]**

Solution:

90. Rewrite the following statement in if-then form. "Either you get to work on time or you are fired" **[2 marks]**

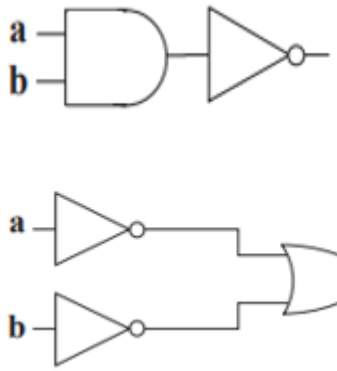
Solution:

91. For the given truth table, form a Boolean function and simplify it. **[6 marks]**

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Solution:

92. Determine whether the following two combinatorial circuits equivalent. **[3 marks]**.



Solution:

93. Show that the given statement is a tautology:

$[(p \Rightarrow q) \wedge p] \Rightarrow q$ **[3 marks]**

Solution:

p	q	$p \Rightarrow q$	$[(p \Rightarrow q) \wedge p]$	$[(p \Rightarrow q) \wedge p] \Rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T
		✓	✓	✓

The final column evaluates to true for all possible combinations of truth values of p and q. Therefore, the given statement is a tautology. ✓

94. Show that the given statement is a tautology:

$[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$ **[4 marks]**

Solution:

p	q	r	$(p \Rightarrow q)$	$(q \Rightarrow r)$	$[(p \Rightarrow q) \wedge (q \Rightarrow r)]$	$(p \Rightarrow r)$	$[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T
			✓		✓		✓

The final column evaluates to true for all possible combinations of truth values of p and q. Therefore, the given statement is a tautology. ✓

95. Construct a truth table and use it to determine whether $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ ia a tautology. **[4 marks]**

Solution: It is a tautology . ✓

				✓	✓	✓
p	q	$\sim p$	$\sim q$	$p \to q$	$\sim q \to \sim p$	$(p \to q) \leftrightarrow (\sim q \to \sim p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

96. Determine whether the statements $(\ p \vee q\ ) \to r$ and $(\ p \to r\ ) \vee (\ q \to r\ )$ logically equivalent?. **[5 marks]**

Solution: They are not logically equivalent . ✓

			✓	✓	✓		✓
p	q	r	$p \vee q$	$(p \vee q) \to r$	$p \to r$	$q \to r$	$(p \to r) \vee (q \to r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

97. Verify that $P \wedge (\sim P + Q') \equiv (P \to Q)'$ using truth table. **[5 marks]**

Solution:

P	Q	$\sim P$	Q'	$\sim P + Q'$✓	$P \wedge (\sim P + Q')$✓	$P \to Q$✓	$(P \to Q)'$✓
1	1	0	0	0	0	1	0
1	0	0	1	1	1	0	1
0	1	1	0	1	0	1	0
0	0	1	1	1	0	1	0

From the truth table, we can observe that the values in the columns $P \wedge (\sim P + Q')$ and $(P \to Q)'$ are the same for all possible combinations of truth values of P and Q. Therefore, the given expression $P \wedge (\sim P + Q') \equiv (P \to Q)'$ holds true. ✓

98. Negate the following quantified statements:

a) Every polynomial function is continuous. **[1 mark]**

Solution: There exists a polynomial that is not continuous everywhere. ✓

b) Some real numbers are rational. **[1 mark]**

Solution: Every real number is irrational ✓

c) Every student has at least one course where the lecturer is a professor. **[1 mark]**

Solution: There is a student such that in every course the lecturer is not a professor ✓

99. What kind of error does the following invalid argument exhibit? Explain.

a) All cheaters sit in the back row. George sits in the back row. $\therefore$ George is a cheater. **[2 marks]**

Solution: The argument is of the form:

$\forall\ x, P(x) \to\ Q(x)$,

$Q(a)$ for some particular a

$\therefore\ P(a)$ ✓

This exhibits universal converse error ✓

b) All honest people pay their taxes.Darth is not honest. $\therefore$ Darth does not pay his taxes.**[2 marks]**

Solution: The argument is of the form:

$\forall\ x, P(x) \to\ Q(x)$,

$\sim P(a)$ for some particular a

$\therefore \sim Q(a)$ ✓

This exhibits universal inverse error ✓

100. Determine whether each the following arguments are valid. Do not use tables.

a) If John drinks beer, he is at least 18 years old. John does not drink beer. Therefore, John is not yet 18 years old. **[3 marks]**

Solution: The argument is of the form:

$p \rightarrow q$,

$\sim p$

$\therefore \sim q$ ✓

This argument exhibits inverse error, hence invalid ✓✓

b) If girls are blonde, then they are popular with boys. Ugly girls are unpopular with boys.

Intellectual girls are ugly. Therefore, blonde girls are not intellectual. **[4 marks]**

Solution: The argument is of the form:

$b \rightarrow p$.....(1),

$u \rightarrow \sim p$.....(2) ✓

$t \rightarrow u$.....(3)

$\therefore b \rightarrow t$

Contrapositive of (2) gives us $p \rightarrow \sim u$.....(4) ✓

Apply transitivity rule on (1) and (4) to get $b \rightarrow \sim u$.....(5), ✓

Contrapositive of (3) gives us $\sim u \rightarrow \sim t$.....(6)

Apply transitivity rule on (5) and (6) to get $b \rightarrow \sim t$.....(5), as required. ✓

101. Derive the truth table for the Sum Of Product expressions

$\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$ **[3 marks]**

Solution:

x	y	z	F	Minterms
1	1	1	1	xyz
1	1	0	0	
1	0	1	0	
1	0	0	0	
0	1	1	1	$\bar{x}yz$
0	1	0	1	$\bar{x}y\bar{z}$
0	0	1	1	$\bar{x}\bar{y}z$
0	0	0	1	$\bar{x}\bar{y}\bar{z}$

✓✓✓

102. Simplify the boolean expression $\bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + \bar{P}QR + PQR$. **[3 marks]**

Solution:

$$\bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + \bar{P}QR + PQR = \bar{P}\bar{Q}(\bar{R} + R) + \bar{P}Q(\bar{R} + R) + PQR \quad ✓$$

$$= \bar{P}\bar{Q}(1) + \bar{P}Q(1) + PQR = \bar{P}\bar{Q} + \bar{P}Q + PQR \quad ✓$$

$$= \bar{P}(\bar{Q} + Q) + PQR$$

$$= \bar{P}(1) + PQR \quad ✓$$

$$= \bar{P} + PQR$$

103. Draw the digital circuit corresponding to the Boolean expression $\bar{a}b + b\bar{c}$. **[3 marks]**

104. Rewrite each of the following statements in the form: " ∀.... if .... then ....":

a) All COBOL programs have at least 20 lines. **[1 mark]**

Solution: ∀ COBOL programs, if the program exists, then it has at least 20 lines. ✓

b) The sum of any two even integers is even. **[1 mark]**

Solution: ∀ even integers a and b, if a and b are even integers, then the sum of a and b is even. ✓

105. Determine whether the following arguments are valid.

a) If I run I will get there quicker.I got there quicker.

Therefore I must have ran. **[3 marks]**

Solution:

✓

b) If I am wealthy, then I am happy. I am happy.

Therefore, I am wealthy. **[3 marks]**

Solution:

✓

c) If John drinks beer, he is at least 18 years old. John does not drink beer. Therefore,

John is not yet 18 years old. **[3 marks]**

Solution: The argument is of the form:

$p \rightarrow q$,

$\sim p$

$\therefore \sim q$ ✓

This argument exhibits inverse error, hence invalid ✓✓

d) If girls are blonde, then they are popular with boys. Ugly girls are unpopular with boys. Intellectual girls are ugly.

Therefore, blonde girls are not intellectual. **[4 marks]**

Solution: The argument is of the form:

$b \rightarrow p$.....(1),

$u \rightarrow \sim p$.....(2) ✓

$t \rightarrow u$.....(3)

$\therefore b \rightarrow t$

Contrapositive of (2) gives us $p \rightarrow \sim u$.....(4) ✓

Apply transitivity rule on (1) and (4) to get $b \rightarrow \sim u$.....(5), ✓

Contrapositive of (3) gives us $\sim u \rightarrow \sim t$.....(6)

Apply transitivity rule on (5) and (6) to get $b \rightarrow \sim t$.....(5), as required. ✓

106. Derive the truth table for the Sum Of Product expressions

$\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$ **[3 marks]**

Solution:

x	y	z	F	Minterms
1	1	1	1	xyz
1	1	0	0	
1	0	1	0	
1	0	0	0	
0	1	1	1	$\bar{x}yz$
0	1	0	1	$\bar{x}y\bar{z}$
0	0	1	1	$\bar{x}\bar{y}z$
0	0	0	1	$\bar{x}\bar{y}\bar{z}$

✓✓✓

107. Simplify the boolean expression $\bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + \bar{P}QR + PQR$. **[3 marks]**

Solution:

$$\begin{aligned}
\bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + \bar{P}QR + PQR &= \bar{P}\bar{Q}(\bar{R} + R) + \bar{P}Q(\bar{R} + R) + PQR \quad \checkmark \\
&= \bar{P}\bar{Q}(1) + \bar{P}Q(1) + PQR = \bar{P}\bar{Q} + \bar{P}Q + PQR \quad \checkmark \\
&= \bar{P}(\bar{Q} + Q) + PQR \\
&= \bar{P}(1) + PQR \quad \checkmark \\
&= \bar{P} + PQR
\end{aligned}$$

108. Draw the digital circuit corresponding to the Boolean expression $\bar{a}b + b\bar{c}$. **[3 marks]**

Solution: ✓

109. An island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B. What are A and B if A says "The two of us are both knights" and B says "A is a knave." **[4 marks]**

Solution:

		A says "The two of us are both knights"		B says "A is a knave."	
A	B	A	B	A	B
Knights	Knights	Knights	Knights	Knaves	Knights
Knights	Knaves	Knights	Knights	Knights	Knaves
Knaves	Knights	Knaves	Knights	Knaves	Knights
Knaves	Knaves	Knaves	Knights	Knights	Knaves

Table or other method ✓✓

A is a Knave while B is a Knight ✓✓

110. Use a truth table to determine whether the argument below is valid.

$p,\ p \to q,\ \sim q \vee r\ \ \therefore\ r$ **[5 marks]**

Solution: .

p	**q**	**r**	$\sim q$	p	$p \to q$	$\sim q \vee r$	r
T	T	T	F	T	T	T	T
T	T	F	F	T	T	F	F
T	F	T	T	T	F	T	T
T	F	F	T	T	F	T	F
F	T	T	F	F	T	T	T
F	T	F	F	F	T	F	F
F	F	T	T	F	T	T	T
F	F	F	T	F	T	T	F
				✓	✓	✓	

Row 1 is a critical row, it has a true conclusion and hence the argument is valid. ✓✓

111. Are these system specifications consistent?

"Whenever the system software is being upgraded, users cannot access the file system.

If users can access the file system,then they can save new files.

If users cannot save new files, then the system software is not being upgraded." **[5 marks]**

Solution:

Let the following statements be represented symbolically as shown:

u: "The software system is being upgraded."

a: "Users can access the file system."

s: "Users can save new files."

Write each system specification symbolically.

"Whenever the system software is being upgraded, users cannot access the file system." $u \rightarrow \sim a$ ✓

"If users can access the file system, then they can save new files." $a \rightarrow s$ ✓

"If users cannot save new files, then the system software is not being upgraded." $\sim s \rightarrow \sim u$ or $u \rightarrow s$ ✓

u	a	s	$\sim u$	$\sim a$	$\sim s$	$u \rightarrow \sim a$	$a \rightarrow s$	$\sim s \rightarrow \sim u$
T	T	T	F	F	F	F	T	T
T	T	F	F	F	T	F	F	F
T	F	T	F	T	F	T	T	T
T	F	F	F	T	T	T	T	F
F	T	T	T	F	F	T	T	T
F	T	F	T	F	T	T	F	T
F	F	T	T	T	F	T	T	T
F	F	F	T	T	T	T	T	T

The system is consistent , for example making u false, a false, and s true makes it consistent. ✓ ✓

112. List any four valid rules of inference for quantified statements **[4 marks]**

Solution:

i) Universal Generalization

ii) Universal Instantiation

iii) Existential Generalization

iv) Existential instantiation

v) Universal Modus Ponens

vi) Universal Modus Tollens

vii) Universal Transitivity

113. Write each of these statements in the form " if p, then q " in English.

i) "For it to be a square, it is necessary that it has four sides" **[1 mark]**

Solution: If it is a square, then it is four sided. ✓

ii) A sufficient condition for the warranty to be good is that you bought the computer less than a year ago. **[1 mark]**

Solution: If you bought the computer less than a year ago then the warranty is good. ✓

iii) Willy gets caught whenever he cheats. **[1 mark]**

Solution: If Willy cheats then he is caught. ✓

114. Determine the truth value for each simple statement. Then using the truth values obtained, determine the truth value of the compound statement:

"Wednesday follows Tuesday or 1 is a prime number, if and only if $p \vee \sim p$ is a contradiction". **[5 marks]**

Solution:

Wednesday follows Tuesday: T ✓

1 is a prime number: F ✓

Wednesday follows Tuesday or 1 is a prime number: T ✓

$p \vee \sim p$ is a contradiction. F ✓

Wednesday follows Tuesday or 1 is a prime number, if and only if $p \vee \sim p$ is a contradiction. : F ✓

115. Write down the converse and the Contrapositive of the implication:

If n is an odd integer, then $n^2 + n - 2$ is an even integer. **[2 marks]**

Solution:

Converse: If $n^2 + n - 2$ is an even integer, then n is an odd integer ✓

Contrapositive: If $n^2 + n - 2$ is an odd integer, then n is an even integer ✓

116. Given:

- Predicate $I(x) : x$ is wise
- Predicate $E(x, y) : x$ knows y
- Domain $P = \{a, b, c\}$
- The truth set of $E(x, y)$ is $\{(a, a), (b, b), (c, c), (a, c), (a, b)\}$

Answer the following questions;

i) Is the statement $(\forall x \in P)(\exists y \in P) : I(y) \rightarrow E(x, x)$ true? Explain **[1 mark]**

Solution: True ✓

$E(x, x)$ is true for all choices of x. ✓

ii) Is the statement $(\exists y \in P) : (\forall x \in P) \sim E(x, y) \vee \sim I(x)$ true?

Explain. **[1 mark]**

Solution: False ✓

$I(a)$ is true so $\sim I(a)$ is false and $\sim E(a, y)$ is false for all choices of y since $E(a, y)$ is true for all $y \in P$

117. Determine whether the expression $(x \rightarrow z) \vee \sim [(x \rightarrow y) \wedge (y \rightarrow z)]$ is a Tautology, Contradiction or a Contingency with help of a truth table. **[4 marks]**

Solution:

x	y	z	$(x \to z)$	$(x \to y)$	$(y \to z)$	$\sim [(x \to y) \wedge (y \to z)]$	$(x \to z) \vee \sim [(x \to y) \wedge (y \to z)]$
T	T	T	T	T	T	F	T
T	T	F	F	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	T	T	T
F	T	T	T	T	T	F	T
F	T	F	T	T	F	T	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T
			✓			✓	✓

The expression evaluates to True for all possible combinations of truth values of x, y, and z.

Therefore, the given expression$(x \to z) \vee \sim [(x \to y) \wedge (y \to z)]$ is a tautology.✓

118. Determine whether the propositional form $(p \to q) \wedge \ (p \wedge \ \sim \ q)$ is a tautology, contradiction or a contingency. **[4 marks]**

Solution:

p	**q**	$\sim$ q	p $\to$ q	$p \wedge \sim q$	$(p \to q) \wedge \ (p \wedge \ \sim \ q)$
T	T	F	T	F	F
T	F	T	F	T	F
F	T	F	T	F	F
F	F	T	T	F	F
			✓	✓	✓

A Contradiction ✓

119. Using the sentence variables (p and q) and the logic symbols ($\sim, \wedge,$ and $\vee$), rewrite the following symbolically.

i) I am hungry but not thirsty. **[1 mark]**

Solution: $p \wedge \sim q$ ✓

ii) The weather is neither hot nor cold. **[1 mark]**

Solution: $\sim (p \vee q) \equiv \sim p \wedge \sim q$ ✓

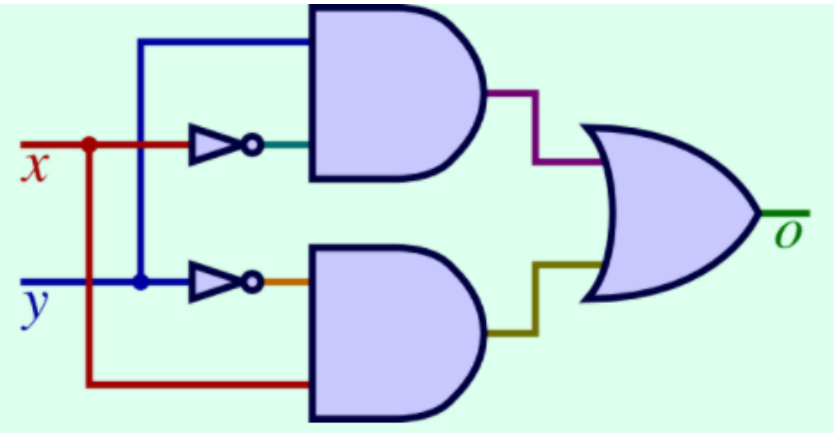
120. Using the Contrapositive proof, show that if n^2 is odd, then n is odd, where n is an integer . **[3 marks]**

Solution: Contrapositive: If n is even, then n^2 is even. ✓

Assume n is even, this implies $n = 2k, k \in \mathbb{Z}$ ✓

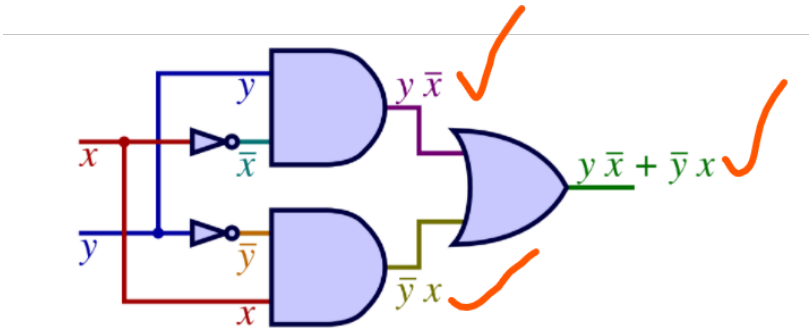
This implies $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ an even number. ✓

121. Find a Boolean expression for the logic circuit below :



[3 marks]

Solution:



122. Using a truth table,determine whether the following argument is valid.

$p \to q$

$q \to r$

$\therefore (p \vee q) \to r$ **[4 marks]**

Solution:

	p	q	r	$p \to q$	$q \to r$	$p \vee q$	$p \vee q \to r$
$\to$	T	T	T	T	T	T	T
	T	T	F	T	F	T	F
	T	F	T	F	T	T	T
	T	F	F	F	T	T	F
$\to$	F	T	T	T	T	T	T
	F	T	F	T	F	T	F
$\to$	F	F	T	T	T	F	T
$\to$	F	F	F	T	T	F	T

All critical rows have a true conclusion and thus the argument is valid ✓

123. For all integers a, b, c, if a divides b and b divides c then a divides c.

Prove **[3 marks]**

Solution:

Since a divides b, then there exists an integer m such that

$b = am$ (i) ✓

Similarly, since b divides c, there exists an integer n such that

$c = bn$ (ii) .

Now, substitute the expression of b from Equation (i) into the b in Equation (ii).

By doing so, the equation $c = bm$ is transformed to $c = (am)n$. ✓

Next, apply the Associative Property of Multiplication on the equation $c = (am)n$ to get $c = a(mn)$. Because m and n are integers, their product must also be an integer by the Closure Property of Multiplication; that is,$m \times n \in \mathbb{Z}$. Let $k = m \times n$. In the equation $c = a(mn)$, substitute mn by k to obtain $c = ak$.The equation $c = ak$ implies that a divides c . ✓

124. Determine a simple logical expression for which each of the following is equivalent to:

i) $\sim (p \rightarrow q)$ **[1 mark]**

Solution: $\sim (p \rightarrow q) \equiv \ p \wedge \ \sim q$ ✓

ii) $\sim (p \iff q)$ **[1 mark]**

Solution: $\sim (p \iff q) \equiv \ p \oplus q$ ✓

iii) $p \wedge \sim p$ **[1 mark]**

Solution: $p \wedge \sim p \equiv c$ where c is a contradiction ✓

iv) $[(p \rightarrow q) \wedge p \ ] \rightarrow q$ **[1 mark]**

Solution: $[(p \rightarrow q) \wedge p \ ] \rightarrow q \equiv t$ where t is a tautology. ✓

125. Use rules of inference to show that if the premises

$\forall x(P(x) \rightarrow Q(x)), \forall x(Q(x) \rightarrow R(x))$, and $\sim R(a)$, where a is in the domain, are true, then the conclusion $\sim P(a)$ is true. **[3 marks]**

Solution:

$\forall x(P(x) \rightarrow Q(x))$

$\forall x(Q(x) \rightarrow R(x))$

$\therefore \forall x(P(x) \rightarrow R(x))$ by universal law of syllogism ✓ ✓

$\forall x(P(x) \rightarrow R(x))$

$\sim R(a)$

$\sim P(a)$ by universal modus tollens ✓

126. Prove the transitivity law : $p \rightarrow q, q \rightarrow r, \therefore \ p \rightarrow r. \ .$ **[6 marks]**

Solution:

p	q	r	$p \to q$	$q \to r$	$p \to r$	Critical rows
T	T	T	T	T	T	*
T	T	F	T	F	F	
T	F	T	F	T	T	
T	F	F	F	T	F	
F	T	T	T	T	T	*
F	T	F	T	F	T	
F	F	T	T	T	T	*
F	F	F	T	T	T	*

✓ ✓ ✓ ✓

All the critical rows are true hence the argument is valid. ✓ ✓

127. If Juan has a smartphone, then she will able to attend the meeting online. When is this statement false? **[1 mark]**

Solution: The statement will be false if Juan has a smartphone but is not able to attend the online meeting. ✓

128. Simplify the Boolean expression: $A + AB$ **[2 marks]**

Solution:

$$\begin{aligned} A + AB &= A(1 + B) \\ &= A.1 \\ &= A \end{aligned}$$

✓✓

129. Prove the following statement :

Root 2 is an irrational number. **[5 marks]**

Solution: Assume that $\sqrt{2}$ is a rational number. ✓

So it can be expressed in the form $\sqrt{2} = \frac{p}{q}$ where p, q are co-prime integers and $q \neq 0$

$\sqrt{2} = \frac{p}{q}$

On squaring both the sides we get,

$2 = \left(\frac{p}{q}\right)^2$

$\Rightarrow \ 2q^2 = p^2$ (i) ✓

$\Rightarrow$ p^2 is even and consequently p is also even.

$\Rightarrow p = 2m, \ m \in \mathbb{Z}$

$\Rightarrow p^2 = 4m^2$ (ii) ✓

From equations (i) and (ii), we get,

$2q^2 = 4m^2$

$\Rightarrow q^2 = 2m^2$

$\Rightarrow q^2$ is a multiple of 2

$\Rightarrow q$ is a multiple of 2 ✓

Hence, p, q have a common factor 2. This contradicts our assumption that they are co-primes.

Therefore, $\frac{p}{q}$ is not a rational number ✓

$\therefore \ \sqrt{2}$ is an irrational number.

130. Let $\Psi = \{p \vee q, \ p \vee \sim q, \ \sim p \vee q, \ \sim p \vee \sim q \vee \sim r, \ \sim p \vee r\}$

Determine whether Ψ is satisfiable. **[5 marks]**

Solution:

p	q	r	$p \vee q$	$p \vee \sim q$	$\sim p \vee q$	$\sim p \vee \sim q \vee \sim r$	$\sim p \vee r$	Ψ
T	T	T	T	T	T	F	T	F
T	T	F	T	T	T	T	F	F
T	F	T	T	T	F	T	T	F
T	F	F	T	T	F	T	F	F
F	T	T	T	F	T	T	T	F
F	T	F	T	F	T	T	T	F
F	F	T	F	T	T	T	T	F
F	F	F	F	T	T	T	F	F

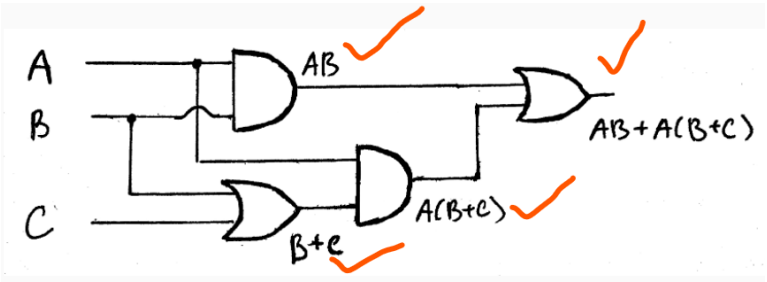
✓ ✓ ✓

There is no assignment where Ψ evaluates to T. ✓

Hence Ψ is un-satisfiable ✓

131. i) Draw the logic circuit for the following expression: $AB + A\,(B + C)$ **[4 marks]**

Solution:



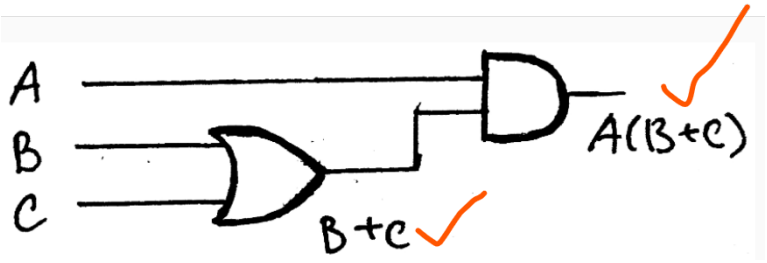
ii) Simplify the expression in i) above using the rules of Boolean algebra. **[2 marks]**

Solution:

$AB + A(B + C) = AB + AB + AC = AB + AC = A(B + C)$ ✓✓

iii) Draw the simplified logic gate circuit derived in ii) above. **[2 marks]**

Solution:



132. You are about to leave for campus in the morning and discover that you don't have your glasses. You know the following statements are true:

a. If I was reading the newspaper in the kitchen, then my glasses are on the kitchen table.

b. If my glasses are on the kitchen table, then I saw them at breakfast.

c. I did not see my glasses at breakfast.

d. I was reading the newspaper in the living room or I was reading the newspaper in the kitchen.

e. If I was reading the newspaper in the living room then my glasses are on the coffee table.

Where are the glasses? **[5 marks]**

Solution Let

RK = I was reading the newspaper in the kitchen.

GK = My glasses are on the kitchen table.

SB = I saw my glasses at breakfast.

RL = I was reading the newspaper in the living room.

GC = My glasses are on the coffee table.

$RK \rightarrow GK$..... by (a)

$GK \rightarrow SB$..... by (d)

$\therefore RK \rightarrow SB$..... by transitivity .....(1) ✓

$RK \rightarrow SB$..... by the conclusion of (1)

$\sim SB$..... by (c)

$\therefore \sim RK$..... by modus tollens .....(2) ✓

$RL \vee RK$..... by (d)

$\sim RK$..... by the conclusion of (2)

$\therefore RL$..... by elimination .....(3) ✓

$RL \rightarrow GC$..... by (e)

RL..... by the conclusion of (3)

$\therefore GC$..... by modus ponens ✓

Thus the glasses are on the coffee table. ✓

133. Find the bitwise OR, bitwise AND, and bitwise XOR of the pair of bit strings: $(110101, 011010)$. **[3 marks]**

Solution:

OR: 1111111 ✓

AND: 010000 ✓

XOR: 101111 ✓

134. Find the bitwise OR, bitwise AND, and bitwise XOR of the pair of bit strings: $(1011110, 0100001)$. **[3 marks]**

Solution:

OR: 1111111 ✓

AND: 0000000 ✓

XOR 1111111 ✓

135. Deduce the conclusion of the argument from the hypothesis explaining each step:

$$p \to q$$

$$q \to r$$

$$\sim r$$

$$\therefore \sim p$$

[2 marks]

Solution:

1.	$q \to r$	Hypothesis
2.	$p \to q$	Hypothesis
3.	$p \to r$	Hypothetical syllogism, 1, 2 ✓
4.	$\neg r$	Hypothesis
5.	$\neg p$	Modus tollens, 3, 4. ✓

136. If x is an even integer, then $x^2 - 6x + 5$ is odd. Prove. **[4 marks]**

Solution:

Suppose x is an even integer.

Then $x = 2a$ for some $a \in \mathbb{Z}$ by definition of an even integer. ✓

so

$$\begin{aligned} x^2 - 6x + 5 &= (2a)^2 - 6(21) + 5 \\ &= 4a^2 - 12a + 5 \\ &= 4a^2 - 12a + 4 + 1 \\ &= 2(2a^2 - 6a + 2) + 1 \end{aligned}$$

✓✓

Therefore, we have $x^2 - 6x + 5 = 2b + 1$, where $b = 2a^2 - 6a + 2 \in \mathbb{Z}$. ✓

Consequently, $x^2 - 6x + 5$ is odd, by definition of an odd number.

137. Show that $1 + 3 + 5 + 7 + ... + (2n - 1) = n^2$ **[4 marks]**

Solution:

Let $P(n) : 1 + 3 + 5 + 7 + ... + (2n - 1) = n^2$

Then $P(1) = 1 = 1^2$ is true. ✓

Assume that $P(k) : 1 + 3 + 5 + 7 + ... + (2k - 1) = k^2$ is true for some $k \in \mathbb{N}$

To prove that $P(k + 1)$ is true we have

$1 + 3 + 5 + 7 + ... + (2k - 1) + (2k + 1) = k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$ ✓✓

Thus $P(k + 1$ is true whenever $P(k)$ is true.

Hence by the principle of mathematical induction , $P(n)$ is true for all $n \in \mathbb{N}$ ✓

138. Determine the validity of the following argument using a truth table

If the apartment is damaged, then the deposit won't be refunded.

The apartment isn't damaged.

Therefore, the deposit will be refunded.

Let p represent "The apartment is damaged."

Let q represent "The deposit will be refunded." **[4 marks]**

Solution:

The argument is of the form:

$p \rightarrow \sim q$

$\sim p$

$\therefore q$ ✓

p	q	$p \rightarrow \sim q$	$\sim p$	q
T	T	F	F	T
T	F	T	F	F
F	T	T	T	T
F	F	T	T	F
		✓		

Critical Rows: row 3 and row 4. ✓

Notice that in the fourth row, both premises are true while the conclusion is false; this "critical row" tells us that the argument is INVALID. ✓

139. Without using truth tables, determine the validity of the argument:

If I go to the movies, I will not do my homework.

I do my homework.

Therefore, I did not go to the movies. **[2 marks]**

Solution: The argument is of the form:

$$p \rightarrow q$$

$$\sim q$$

$$\therefore \sim p$$

✓

This is valid by Modus Tollens. ✓

140. A college cafeteria line has four stations: salads, main courses, desserts, and beverages.

The salad station offers a choice of green salad or fruit salad; the main course station offers spaghetti or fish; the dessert station offers pie or cake; and the beverage station offers milk, soda, or coffee.

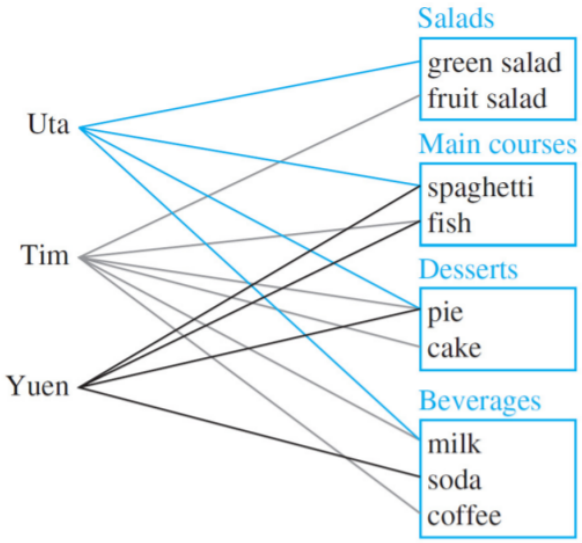
Three students, Uta, Tim, and Yuen, go through the line and make the following choices:

Uta: green salad, spaghetti, pie, milk

Tim: fruit salad, fish, pie, cake, milk, coffee

Yuen: spaghetti, fish, pie, soda.

These choices are illustrated in the figure below:



Write each of following statements informally and find its truth value.

i) $\exists$ an item I such that $\forall$ students x, x chose I . **[2 marks]**

Solution:

There is an item that was chosen by every student. ✓

This is true; every student chose pie. ✓

ii) $\exists$ a student x such that $\forall$ items I, x chose I . **[2 marks]**

Solution:

There is a student who chose every available item. ✓

This is false; no student chose all nine items. ✓

iii) $\exists$ a student x such that $\forall$ stations Z, $\exists$ an item I in Z such that x chose I . **[2 marks]**

Solution:

There is a student who chose at least one item from every station. ✓

This is true; both Uta and Tim chose at least one item from every station. ✓

iv) $\forall$ students x and $\forall$ stations Z, $\exists$ an item I in Z such that x chose I. **[2 marks]**

Solution:

Every student chose at least one item from every station. ✓

This is false; Yuen did not choose a salad. ✓

141. In a bookstore, there are three genres of books: fiction, mystery, and science fiction. The available books in each genre are as follows:

- Fiction: novels or short stories

- Mystery: detective novels or thrillers

- Science Fiction: space adventures or dystopian novels

Three readers, Alex, Brooke, and Chris, make the following choices:

- Alex: Novels, detective novels, space adventures

- Brooke: short stories, thrillers, dystopian novels

- Chris: Novels, detective novels, dystopian novels

Write each of the following statements informally and find its truth value:

i) $\exists$ a book A such that $\forall$ readers R, R chose A . **[2 marks]**

Solution:

Informally: There is a book that was chosen by every reader. ✓

This is False; No book was chosen by every reader. ✓

ii) $\exists$ a reader R such that $\forall$ books B, R chose B. **[2 marks]**

Solution:

Informally: There is a reader who chose every available book. ✓

This is false; no reader chose all books. ✓

iii) $\exists$ a reader R such that $\forall$ genres G, $\exists$ a book B in G such that R chose B . **[2 marks]**

Solution:

Informally: There is a reader who chose at least one book from every genre. ✓

This is true; All the readers chose at least one book from every genre. ✓

iv) $\exists$ a genre G such that $\forall$ readers R, R chose a book from G. **[2 marks]**

Solution:

Informally: There is a genre such that every reader chose a book from that genre.. ✓

This is false; there is no genre chosen by all three readers.. ✓

142. Consider the following truth table. Determine and simplify the logic expression for the output Y above. **[3 marks]**

			Output
A	B	C	Y
1	1	1	0
1	1	0	0
1	0	1	1
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	0

Solution: There are two rows with output 1's and their corresponding binary values are 001 and 101 which correspond to product terms $\bar{A}\bar{B}C$ and $A\bar{B}C$ respectively. ✓

The resulting expression for the output is

$Y = \bar{A}\bar{B}C + A\bar{B}C = (\bar{A} + A)\bar{B}C = 1.\bar{B}C = \bar{B}C.$ ✓✓

143. **[2 marks]**

Solution: ✓✓